

KERCH, Ukraine

WMO: 339830

Lat: 45.3742N

Lon: 36.4147E

Elev: 49

StdP: 100.74

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.0	-13.2	1.2	-10.0	-10.9	1.5	-7.5	14.1	-1.6	12.5	-1.9	5.3	60	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	32.1	22.3	30.8	21.9	29.5	21.4	24.0	29.4	23.2	28.5	22.4	27.6	4.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.1	27.1	21.4	16.2	26.3	20.6	15.4	25.7	72.4	29.3	69.1	28.3	66.2	27.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.2	8.6	DB	-12.8	34.0	3.9	1.7	-15.6	35.2	-17.9	36.2	-20.1	37.1	-22.9	38.3	(4)
(5)			WB	-13.1	25.6	3.7	1.3	-15.8	26.6	-17.9	27.3	-20.0	28.1	-22.8	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.7	1.4	4.8	10.3	16.2	21.0	24.0	23.9	18.6	12.5	7.0	3.0	(6)
(7)		DBStd	9.15	4.95	5.03	3.72	3.25	3.36	2.94	2.53	2.92	3.35	3.97	4.19	4.69	(7)
(8)		HDD10.0	1079	289	243	164	34	1	0	0	0	0	21	107	220	(8)
(9)		HDD18.3	2814	547	476	419	242	83	9	0	1	37	185	339	476	(9)
(10)		CDD10.0	1812	0	1	3	43	194	331	434	431	257	98	18	3	(10)
(11)		CDD18.3	506	0	0	0	0	17	91	176	173	45	4	0	0	(11)
(12)		CDH23.3	4045	0	0	0	2	84	610	1574	1541	226	8	0	0	(12)
(13)		CDH26.7	1198	0	0	0	0	6	134	502	518	37	1	0	0	(13)
(14)	Wind	WSAvg	4.5	4.8	5.1	5.1	4.5	4.0	4.2	4.3	4.3	4.4	4.3	4.8	(14)	
(15)	Precipitation	PrecAvg	404	27	30	28	37	27	51	43	46	38	25	37	40	(15)
(16)		PrecMax	602	69	81	75	110	104	154	143	139	156	82	84	100	(16)
(17)		PrecMin	255	6	12	1	4	0	4	0	0	2	3	0	3	(17)
(18)		PrecStd	83	18	18	19	23	21	38	44	40	38	18	24	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.8	16.1	22.5	27.5	31.6	33.5	34.2	30.1	24.2	18.7	14.7	(19)
(20)			MCWB	9.2	9.4	10.4	14.9	19.6	21.9	22.7	23.1	20.8	18.0	14.7	12.1	(20)
(21)		2%	DB	10.0	10.7	13.6	19.7	25.6	29.6	32.1	32.4	27.5	22.2	16.5	12.6	(21)
(22)			MCWB	8.3	8.4	9.3	13.6	18.7	21.2	22.5	22.3	20.0	17.3	13.7	10.8	(22)
(23)		5%	DB	8.8	9.2	11.7	17.6	24.0	28.2	30.8	31.0	25.8	20.2	14.7	11.2	(23)
(24)			MCWB	7.4	7.3	8.3	12.5	18.0	20.8	22.1	21.8	19.0	16.6	12.6	9.7	(24)
(25)		10%	DB	7.6	7.8	10.1	15.7	22.4	26.8	29.5	29.6	24.3	18.5	13.1	9.8	(25)
(26)			MCWB	6.4	6.4	7.5	11.4	17.1	20.0	21.6	21.3	18.3	15.4	11.3	8.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.0	11.2	16.3	21.2	24.1	25.2	22.8	19.5	15.4	12.8	(27)	
(28)			MCDWB	11.1	11.9	15.4	21.2	26.1	29.1	30.5	30.9	27.7	22.8	17.8	13.8	(28)
(29)		2%	WB	8.5	8.7	9.8	14.2	19.6	22.8	24.1	23.9	21.2	18.2	14.0	10.9	(29)
(30)			MCDWB	9.8	10.4	12.7	18.6	24.2	27.7	29.4	29.5	25.7	20.7	16.1	12.2	(30)
(31)		5%	WB	7.6	7.6	8.8	12.9	18.6	21.7	23.2	23.1	20.2	17.0	12.8	9.8	(31)
(32)			MCDWB	8.8	9.0	11.2	16.9	22.9	26.5	28.6	28.5	24.3	19.6	14.5	11.1	(32)
(33)		10%	WB	6.4	6.4	7.8	11.8	17.6	20.8	22.5	22.4	19.2	15.7	11.5	8.6	(33)
(34)			MCDWB	7.5	7.7	9.8	14.9	21.6	25.3	27.7	27.6	23.0	18.2	13.0	9.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.5	6.5	7.9	8.5	8.3	8.6	8.4	8.2	7.0	5.9	5.0	(35)
(36)			MCDBR	6.3	7.0	9.1	11.0	10.5	9.9	9.8	9.9	10.0	9.3	8.0	6.8	(36)
(37)			MCWBR	5.2	5.1	5.8	6.2	5.4	4.6	4.3	4.1	5.0	5.5	5.6	5.5	(37)
(38)		5% WB	MCDBR	6.3	6.5	8.3	9.8	9.6	8.8	8.9	9.1	9.2	8.6	7.7	6.6	(38)
(39)			MCWBR	5.3	5.0	5.7	5.9	5.5	4.8	4.7	4.5	5.3	5.6	5.7	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.352	0.382	0.411	0.447	0.414	0.418	0.440	0.443	0.414	0.387	0.360	0.351	(40)	
(41)		taud	2.314	2.236	2.179	2.123	2.323	2.353	2.302	2.279	2.323	2.348	2.358	2.315	(41)	
(42)		Ebn at Noon	732	778	809	813	856	852	827	809	801	767	719	687	(42)	
(43)		Edh at Noon	81	105	127	147	124	121	126	124	110	93	78	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.06	1.85	3.17	4.88	6.18	6.67	6.69	5.88	4.30	2.63	1.42	0.92	(44)	
(45)		RadStd	0.14	0.27	0.29	0.45	0.48	0.39	0.30	0.38	0.30	0.27	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.73	N/A	N/A	+1.11	N/A	N/A	-137	-202	+178	+126		
(47) Regional (5 neighbors)		+0.71	N/A	N/A	N/A	N/A	N/A	-118	-171	+156	+93		

Nomenclature: See separate page